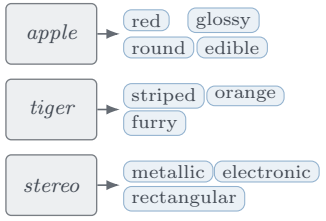
1 Open tagging

Gemma 4 26B, open vocabulary, train split only

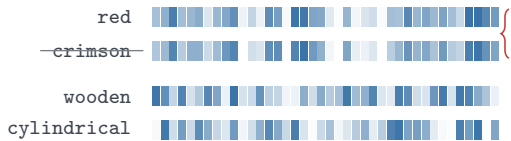


18,211 training images → 1,349 raw phrases

sparse and open: a few tags per image, never a complete judgement

2 Profile and cluster

class-mean image–text score profile over the 1,298 training classes



profile correlation ≥ .5 merges the pair; the more prevalent name be-

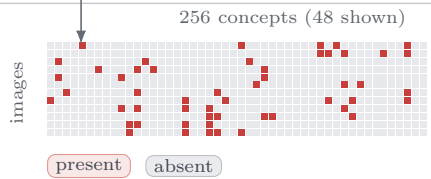
dictionary frozen: 256 concepts

1,349 → 643 candidates (≥3 mentions, ≥3 classes)

→ 303 clusters → top 256 by class-weighted prevalence

3 Exhaustive labelling

closed-set `YES/NO` checklist: every image against all 256 names



≈22.5 of 256 concepts present per image

becomes the label matrix Y of Figure 2